SOFTWARE REQUIREMENTS SPECIFICATION

QUIZ MAKER & STUDY ASSISTANT

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1. INTRODUCTION

1.1 Purpose of the Document

This Software Requirements Specification (SRS) describes what the team expects the Quiz Maker & Study Assistant to do. It outlines the main features, functions, and requirements that the developers plan to implement. This SRS document will assist the team during development and testing to ensure alignment. Once developed, this specification can also help assess whether the application fulfills the agreed-upon requirements.

1.2 Scope of the Project

The Quiz Maker & Study Assistant will be a software application that enables both teachers and students to create quizzes. Teachers will be able to develop quizzes, add questions, and manage quizzes for their classes. Students will also be able to create, add questions to, and manage their own quizzes for studying. Students will be able to take their own study quizzes or quizzes assigned by teachers, receive scores, save quizzes for later, and review their results. Teacher review restrictions apply to teacher-created quizzes; students can review their own study quizzes without teacher permission. Some of the main functionalities to include in this scope are:

- Quiz creation: Teachers and students will develop quizzes and save them to their own quiz lists. Students can create quizzes for studying.
- Quiz completion: Students will take their own study quizzes or quizzes offered by teachers.
- Grading and results: The application evaluates student responses and calculates their grade scores.
- Saved quizzes: Students will store quizzes they wish to revisit in the saved-quizzes section.
- Quiz review: Students will examine their answers, feedback, correctness status, and the correct response for their own study quizzes. For teacher-created quizzes, this review is available when the teacher enables it. The project covers these primary functionalities. Additional study-related functions and complex functionalities will not fall within this scope unless added by the team later. The eventual platform has not yet been determined.

1.3 Stakeholders Involved

1.3.1 Teachers

Teachers will create quizzes for their classes. Teachers need a way to generate quizzes, add questions, and save quizzes to the teacher quiz list. Teachers also need to determine whether their students can access reviews of teacher-created quizzes.

1.3.2 Students

Students will use the software to create their own study quizzes, add questions and correct answers, manage and save their quizzes, take quizzes, and view scores and feedback. Students can review their own study quizzes without teacher permission and review teacher-created quizzes when the teacher permits it.

1.3.3 Development Team

The development team will build and test the quiz maker & study application. The team will refer to this document in order to determine functionality.

1.3.4 Instructor/Project Reviewer

The instructor or project reviewer evaluates the assignment. The reviewer assesses whether the team's requirements meet the assignment expectations.

1.3.5 Schools and Universities

Schools and universities have an interest in how the application supports teaching, learning, and student study. They benefit from teachers creating quizzes for their classes and students creating their own study quizzes. Their interests include usability, accessibility, student privacy, and alignment with institutional educational goals and policies.

1.4 Overview of the Quiz Maker & Study Assistant

1.4.1 Teacher and Student Quiz Creation

Teachers and students use the quiz-creation tool, add questions and correct answers, and save quizzes to their own quiz lists. Students use these quizzes to study and practice. The application needs to clearly direct both user groups through this process.

1.4.2 Student Quiz Completion

Students open their own study quiz or a quiz offered by a teacher, respond to all questions, and submit the quiz attempt. Upon submission, the application evaluates answers, calculates student scores, and returns these scores.

1.4.3 Saved Quizzes

Students can choose to save quizzes in the saved-quizzes section of the application in order to review them in the future. Any duplicated attempts of saved quizzes automatically overwrite previous versions.

1.4.4 Quiz Scores and Results

Students get feedback regarding their performance on quizzes they submit, which includes the calculated score. Students can view answers and feedback for their own study quizzes. For teacher-created quizzes, they can view this information when the teacher grants review access.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The Quiz Maker & Study Assistant will be a piece of software that is currently being developed by the team as part of the project. As such, the platform has not yet been determined, nor have any technical specifications. The software will support both teacher and student roles, and both roles can create and manage their own quizzes. Teachers create quizzes for their classes, while students can create quizzes for personal study. Teacher review restrictions apply to teacher-created quizzes and do not prevent students from reviewing their own study quizzes.

2.2 Product Functions

The system is expected to fulfill the following requirements:

- Enable teachers and students to create quizzes and add questions and correct answers. Each quiz will be saved to its creator’s quiz list. Students can use their own quizzes for studying.
- Permit a student to select and submit a quiz attempt.
- Automatically evaluate quiz responses and calculate student scores.
- Enable a student to save quizzes and review quizzes saved in the past. When saved again, duplicates should not be stored.
- Allow students to review their own study quizzes without teacher permission. For teacher-created quizzes, apply the teacher's review settings.

• Restrict a student from reviewing results of a teacher-created quiz when that teacher has blocked review access.

2.3 User Characteristics

2.3.1 Teachers

Teachers should be able to navigate the system interface, create quizzes, enter questions, save quizzes, and control review access for quizzes they create.

2.3.2 Students

The quiz creation and quiz taking processes should be intuitive enough for students to create and save their own study quizzes, enter questions and correct answers, select quizzes, answer questions, submit their responses, and understand the feedback received upon completion.

2.3.3 Development Team

The development team will use the requirements outlined in this document when implementing the software. The development team will select a platform for this application and design the technical solution based on this choice.

2.4 Constraints

• The application should calculate a student's score immediately upon submission of their attempt and display the score within three seconds.
• Teacher permission is required to access feedback on teacher-created quizzes. Students can access feedback on their own study quizzes without teacher permission.
• The software is required to satisfy the functions described above in relation to both teachers and students. The specific technical architecture, the eventual platform, and other implementation decisions are undetermined at present.

2.5 Assumptions and Dependencies

• Teachers will create quizzes and make them accessible to students. Students will also be able to create quizzes for their own study use.
• Students will have access to their own study quizzes and to quizzes made available by their teachers.
• Question texts and possible response options will be entered correctly in the application. Question formats have not yet been specified by the team.
• Quiz evaluation requires correct answers entered into the application to compare submitted student responses.
• Review access for teacher-created quizzes depends on the teacher's settings. Review of a student's own study quizzes does not require teacher permission.
• Both teachers and students will have access to suitable environments for using the software. The particular platform has not yet been determined.

2.6 Apportioning of Requirements

The current version of the software includes basic features outlined above in the product description. Advanced features such as quiz analytics, recommendations, integration with the learning management system, and any other features have not been defined within the scope of this version. The team reserves the right to define any additional features in future updates.

3. SPECIFIC REQUIREMENTS

In this section, the team defines specific functionality and performance requirements that apply to the Quiz Maker & Study Application. All functional requirements relate to the actions the system must perform. All non-functional requirements relate to the desired qualities of the application's performance.

3.1 Functional Requirements

3.1.1 Quiz Creation

FR 1 — Teacher or Student Creates a Quiz

The system should allow both teachers and students to create quizzes, add questions and correct answers, and save quizzes to their personal quiz lists. Students must be able to create quizzes for their own study use.

Acceptance Criteria

- Teachers and students will use buttons and fields provided by the application to create quizzes.
- After entering all necessary data regarding the quiz questions and answers, the teacher or student should use the button to submit quiz information to the system.
- The system saves the newly created quiz in its creator's quiz list, including a student's own study quizzes.

3.1.2 Taking Quizzes and Grading

FR 2 — Student Takes a Quiz and Gets the Score

The application should allow students to answer quiz questions and submit their attempts. After submission, the system automatically grades student responses and reports their scores.

Acceptance Criteria

- Students will open their own study quizzes or quizzes provided by teachers. Students will use buttons provided by the system to respond to quiz questions.
- Students will use a button to submit their answers.
- The system evaluates submitted student attempts. Submitted responses will be evaluated based on the pre-determined criteria in order to generate student scores. Note: The question formats and scoring criteria will be further detailed by the team.

3.1.3 Saving Quizzes

FR 3 — Student Saves a Quiz

The system should enable students to save quiz attempts in their personal saved-quizzes folders.

Acceptance Criteria

- Students use a designated button to initiate a request to save the attempted quiz.
- The system checks whether the particular quiz attempt has already been saved before storing a new quiz attempt in the student's saved-quizzes folder. If stored before, then the previously stored quiz attempt will be replaced by a newer attempt.
- The application lists all of the quizzes saved by a student in the saved-quizzes folder upon opening the folder.

3.1.4 Reviewing Quiz Results

FR 4 — Student Reviews a Quiz

After submitting a quiz, the application should allow students to review their responses and results for their own study quizzes without teacher permission. For teacher-created quizzes, review is available only when enabled by the teacher.

Acceptance Criteria

- The application provides a link or button labelled 'Review', which becomes visible after a student has finished and submitted his/her quiz responses.
- When clicking on 'Review,' students will access a page that displays all questions along with their response. Students can see whether they got a question right or wrong. Correct answers will also be provided alongside the questions answered incorrectly by the student.
- For a teacher-created quiz that does not permit review access, students receive appropriate notification. If the student tries to access quiz review results, he/she will see corresponding message indicating inability to view quiz feedback.

3.2 Non-Functional Requirements

This set of requirements addresses various aspects of system quality. Requirements related to the actual implementation, including the eventual platform, are stated generally until the decision on the implementation details is made by the team.

3.2.1 Performance Requirements

NFR 1 — Quiz Results Calculation Time

The system should process the submitted quiz and calculate the student's score within three seconds.

Acceptance Criteria: Within testing conditions used during the project, the system response time from receiving the submission to reporting the grade will be no greater than three seconds.

3.2.2 Usability Requirements

NFR 2 — User-Friendly Navigation

Navigation through the application must be clear for the two user groups involved. Navigation should be easy for teachers to locate and activate quiz creation and storage functions. Similarly, navigation must be user-friendly for students creating their own study quizzes or looking for quizzes, saved quizzes, quiz results, and quiz reviews.

3.2.3 Security and Access Control Requirements

NFR 4 — Function Permissions

The application must allow both teachers and students to create and manage their own quizzes. Students must not be prevented from using quiz-creation functions for personal study. Users may modify only quizzes they own or are authorized to edit. The application must enforce teacher review restrictions on teacher-created quizzes while allowing students to review their own study quizzes without teacher permission.

3.2.5 Availability and Reliability Requirements

NFR 7 — Consistency and Accuracy of Quiz Storage

The application should provide confirmation that quiz data was stored. Failure to perform any actions should be communicated to users. False confirmation messages are not acceptable in this application. The requirement does not include requirements regarding uptime, availability times per day, or other similar technical parameters for availability purposes because these specifics have not been determined for the class project.

4. USE CASES

The following use cases illustrate the interaction between users and the Quiz Maker & Study Assistant. Each use case is explained in accordance with templates presented in the SRS sample provided by the instructor.

4.1 Teacher or Student Creates a Quiz

Description: A teacher or student creates a quiz, enters questions and correct answers, and saves the completed quiz to their own quiz list. Students can create quizzes for studying.

Actors: Teacher or Student – the user performing quiz creation activities. System – the Quiz Maker & Study Assistant application.

Pre-condition: The teacher or student has access to the application and its quiz-creation features.

Post-condition: The quiz and its entered questions and correct answers are saved to the creator's quiz list. A student-created quiz is available for that student's study use.

Basic Flow:

1. Activate quiz creation tool.
2. Enter quiz questions and other relevant information.
3. Save quizzes to the quiz storage area.

Alternative flows: If necessary, teachers or students may add multiple questions.

Exception: Technical failures may occur during quiz creation and storage.

4.2 Student Takes a Quiz

Description: Attempting an available quiz, answering questions, submitting answers, and receiving grades.

Actors: Student – the individual participating in quizzes. System – the application offering quizzes to students, grading quiz attempts, and showing quiz grades.

Pre-condition: Students have access to the quiz-taking application. At least one available quiz exists.

Post-condition: Quiz answers are evaluated and the resulting grade is displayed.

Basic Flow:

7. Open your own study quiz or a quiz offered by your teacher.
8. See all quiz questions.
9. Type in answers.
10. Click Submit after finishing the quiz.
11. Application evaluates answers submitted.
12. Grade for this quiz attempt is reported.

Alternative flows: Students may notice missing answer fields; the application should recognize empty spaces. Students might try to leave quiz midway without submitting it.

Exceptions: The system cannot evaluate submitted answers for technical reasons; it notifies the user of unsuccessful submission and attempts at processing.

4.3 Student Saves a Quiz

Description: Saving an attempted quiz so students could review their submissions later.

Actors: Student – the user initiating the attempt-saving activity. System – the application storing attempted quizzes in memory.

Pre-condition: Students access the quiz-taking application. The student has opened a quiz attemptable by him/her.

Post-condition: The quiz attempt gets stored in memory.

Basic Flow:

13. Request saving the quiz attempt from the system.
14. Check the system records for this attempt's existence in the save storage space.
15. If stored previously, overwrite the earlier attempt record with the newer attempt record.
16. Otherwise, store the attempt.
17. Confirm successful saving attempt.
18. Students access quiz storage page from the main menu.
19. View quiz storage listing.

Alternative flows: Overwriting occurs automatically. Quiz storage will remain even if the student does not open the saved quizzes folder immediately.

Exceptions: Attempt storage fails due to a problem. An error message should appear.

4.4 Student Reviews Quiz Results

Description: Viewing questions, answers, evaluation information, and correct answers when available.

Actors: Student – the individual attempting quizzes. System – the program evaluating quiz attempts and allowing review access. Teacher – enabling or blocking reviews of teacher-created quizzes.

Pre-condition: The student has attempted the quiz. The quiz attempt was graded by the application. The quiz is the student's own study quiz, or the teacher has enabled review of the teacher-created quiz.

Post-condition: The student can see the list of questions, answers, evaluation information, and correct answers.

Basic Flow:

20. Initiate the review activity by activating the 'review' button.
21. Allow review of the student's own study quiz; for a teacher-created quiz, check whether the teacher has enabled review.
22. Retrieve the student's quiz attempt results and associated quiz questions and answers.
23. Display questions and submitted answers one-by-one.
24. Show each answer's correctness status.
25. Display correct answer for every incorrect response.

Alternative flows: Teachers can disable review of teacher-created quizzes, and students receive notifications about this. Teacher review restrictions do not apply to a student's own study quizzes.

Exceptions: Technical errors prevent retrieving saved information. The system informs students about this failure.

5. APPENDICES

This appendix contains definitions of important terminology, abbreviations, acronyms, references, and resources relevant to this project.

5.1 GLOSSARY OF TERMS

Acceptance criterion – A condition that must be fulfilled in order to call a requirement fulfilled.

Automatic grading – Evaluating submitted answers against predefined answers. Functional Requirement (FR) – A requirement specifying action performed by the application. Non-Functional Requirement (NFR) – A requirement defining some quality or property of an application. Quiz review – An ability to revisit submitted quizzes and view results, evaluations, and correct answers. Saved quizzes folder – A place to store submitted quizzes. Stakeholder – Anyone who has a stake in this project. Use case – A scenario involving users and the system.

5.2 ACRONYMS AND ABBREVIATIONS

SRS – Software Requirements Specification FR – Functional Requirement NFR – Non-Functional Requirement UI – User Interface

5.3 REFERENCES AND RESOURCES

Case Study – File 4 – Sample SRS Doc.pdf, provided as format and organizational guideline.